

## Design Considerations

### Construction

Open frame solenoids are designed with two frame styles, the C Frame style, in which the coil is enclosed on one side, and the Box Frame style in which the coil is enclosed on two sides. The Box Frame style provides slightly higher force output and is more rugged in design.

Tapped mounting holes are used for easy installation and interchangeability.

Most models have slotted and cross drilled plungers for easy load attachment.

The plunger is plated for corrosion resistance, and provides a low coefficient of friction and long life.

Over molded coils are available in both Box Frame and C Frame solenoids and offer excellent protection from moisture and humidity. Some solenoids are UL recognized. Most have UL recognized coil insulation systems.

### Life

When selecting an open frame solenoid, as with any other solenoid style, it is important to consider the effects of heat, since an increase in coil temperature reduces the work output and the life of the unit. Life ratings extend to 3 million cycles depending on the product size and application. Consult the factory for longer life of 500,000 or more cycles, and other special requirements.

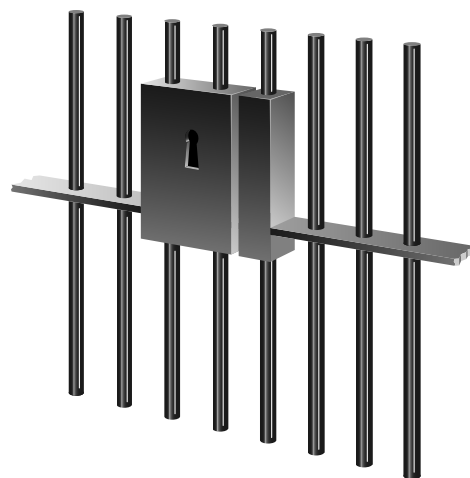
### Duty Cycle

Duty cycle is determined by solenoid ON time/(ON + OFF time).

For example: a solenoid is actuated for 30 seconds, then off for 90 seconds.  
 $30 \text{ sec ON} / (30 \text{ Sec ON} + 90 \text{ sec OFF}) = 30/120 = 1/4$   
or 25% duty cycle.

### Performance Curves

The Force/Stroke performance curves in this section serve as guides to determine the solenoid size needed to produce a desired force at a given stroke, duty cycle, and power source. All Force/Stroke curves are performed under standard test conditions: ambient temperature of 20°C. A design safety factor of 1.3 to 1.5 is recommended. For example, when a 4.5 lb pull force is required, select a model with a safety factor of 1.3 to 1.5 times (5.8 to 6.7 lb).



# Ledex® Box Frame Size B12

Part Number: B12 - **XXX** - B-3

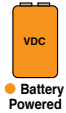
All products are RoHS Compliant

LINEAR Open Frame

Select from performance chart below

## Specifications

|                         |                                  |
|-------------------------|----------------------------------|
| Operation               | Pull                             |
| Dielectric Strength     | 500 VRMS for one second          |
| Continuous Duty Cycle   | 100% at 20°C ambient temperature |
| Intermittent Duty Cycle | See below                        |
| Holding Force           | 7.7 oz. (2.1 N) at 20°C          |
| Coil Insulation         | Class "A": 105°C max.            |
| Coil Termination        | 10" (254 mm) PVC lead wires      |
| Plunger Weight          | 0.05 oz. (1.4 g)                 |
| Total Weight            | 0.30 oz. (8.5 g)                 |



## Performance

|                                                   |      |     |     |     |
|---------------------------------------------------|------|-----|-----|-----|
| Maximum Duty Cycle                                | 100% | 50% | 25% | 10% |
| Maximum ON Time (sec)<br>when pulsed continuously | ∞    | 84  | 11  | 2   |
| Maximum ON Time (sec)<br>for single pulse         | ∞    | 120 | 34  | 8   |
| Watts (@ 20°C)                                    | 1.3  | 2.6 | 5.2 | 13  |
| Ampere Turns (@ 20°C)                             | 178  | 251 | 355 | 561 |

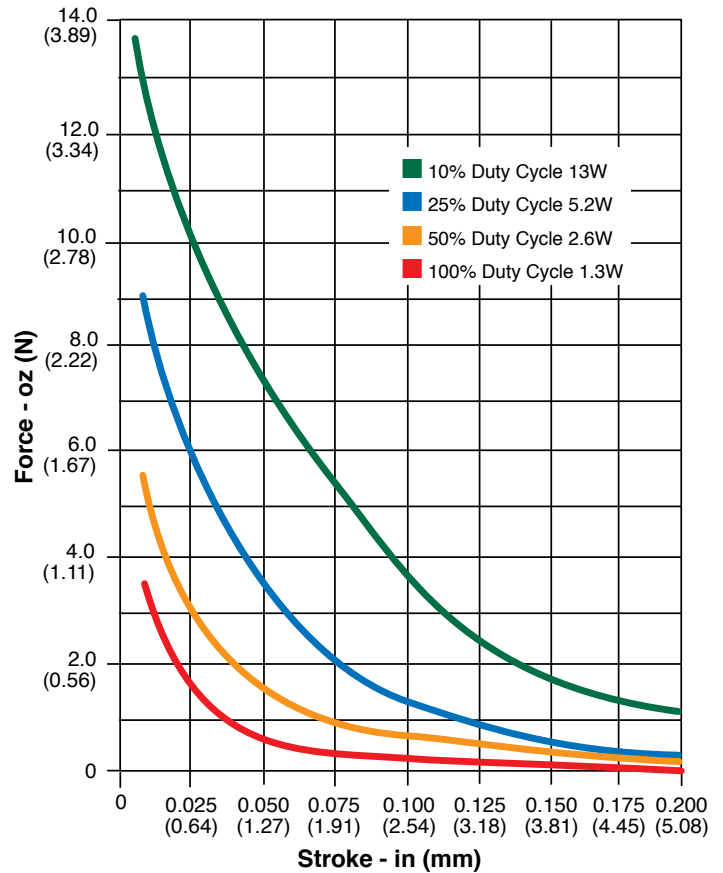
## Coil Data

| Part Number | Resistance<br>(@20°C) | Ref #<br>Turns | VDC<br>(Nom) | VDC<br>(Nom) | VDC<br>(Nom) | VDC<br>(Nom) |
|-------------|-----------------------|----------------|--------------|--------------|--------------|--------------|
| B12-258-B-3 | 6.92                  | 417            | 3            | 4.24         | 6            | 9.5          |
| B12-255-B-3 | 27.7                  | 824            | 6            | 8.5          | 12           | 19           |
| B12-256-B-3 | 62.3                  | 1184           | 9            | 13           | 18           | 28.5         |
| B12-254-B-3 | 110.8                 | 1632           | 12           | 17           | 24           | 38           |
| B12-253-B-3 | 443.1                 | 3336           | 24           | 34           | 48           | 76           |

## NOTES:

1. All data is typical.
2. Force testing is done with the solenoid in the horizontal position.
3. All data reflects operation with no heat sink.
4. Pull versions standard; push versions available.
5. Magnetic latching version available.

## Typical Force @ 20°C



## How to Order

Select the part number from the table provided. (For example, to order a 25% duty cycle unit rated at 48 VDC, specify B12-253-B-3.

Please see [www.ledex.com](http://www.ledex.com) for our list of stock products available through our North American distributors.

All specifications subject to change without notice.

Force values for reference only.